

Pilot study to explore effects of low-fat, flaxseed-supplemented diet on proliferation of benign prostatic epithelium and prostate-specific antigen.

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OBJECTIVES: Dietary factors may influence the prostate and have an impact on prostatic growth and disease. A small number of studies have suggested that flaxseed-supplemented, fat-restricted diets may thwart prostate cancer growth in both animals and humans. Unknown, however, is the potential effect of such a diet on benign prostatic epithelium. METHODS: We undertook a pilot study to explore whether a flaxseed-supplemented, fat-restricted diet affects the proliferation rates in benign epithelium. We also explored the effects on circulating levels of prostate-specific antigen (PSA), total testosterone, and cholesterol. Fifteen men who were scheduled to undergo repeat prostate biopsy were instructed to follow a low-fat (less than 20% kcal), flaxseed-supplemented (30 g/day) diet and were provided with a supply of flaxseed to last throughout the 6-month intervention period. The PSA, total testosterone, and cholesterol levels were determined at baseline and at 6 months of follow-up. Reports from the original and repeat biopsies were compared, and proliferation (MIB-1) rates were quantified in the benign prostatic epithelium. RESULTS: Statistically significant decreases in PSA (8.47 ± 3.82 to 5.72 ± 3.16 ng/mL; $P = 0.0002$) and cholesterol (241.1 ± 30.8 to 213.3 ± 51.2 mg/dL; $P = 0.012$) were observed. No statistically significant change was seen in total testosterone (434.5 ± 143.6 to 428.3 ± 92.5 ng/dL). Although 6-month repeat biopsies were not performed in 2 cases because of PSA normalization, of the 13 men who underwent repeat biopsy, the proliferation rates in the benign epithelium decreased significantly from 0.022 ± 0.027 at baseline to 0.007 ± 0.014 at 6 months of follow-up ($P = 0.0168$). CONCLUSIONS: These pilot data suggest that a flaxseed-supplemented, fat-restricted diet may affect the biology of the prostate and associated biomarkers. A randomized controlled trial is needed to determine whether flaxseed supplementation, a low-fat diet, or a combination of the two regimens may be of use in controlling overall prostatic growth.